

Kali Linux Command

BY

Intern Name :-Shruti Rane

Intern Id:-2066

1. **ls** :- Lists files and folders in the current directory

```
(shrutirane@shrutirane) [-]  
$ ls  
Desktop Documents Downloads Music Pictures Public Templates Videos
```

2. **pwd** :- Shows the current working directory

```
(shrutirane@shrutirane) [-]  
$ pwd  
/home/shrutirane
```

3. **cd**:- Changes the current directory

```
(shrutirane@shrutirane) [-]  
$ ls  
Desktop Documents Downloads Music Pictures Public Templates Videos  
  
(shrutirane@shrutirane) [-]  
$ cd Desktop
```

4. **find** :- Searches for files and directories in a specified location

```
(shrutirane@shrutirane) [-~/Desktop]  
$ find s12  
s12  
  
(shrutirane@shrutirane) [-~/Desktop]  
$
```

5. **man** :- Displays the user manual for a command

```
shrutirane@shrutirane -  
Session Actions Edit View Help  
TOUCH(1) User Commands TOUCH(1)  
  
NAME  
touch - change file timestamps  
  
SYNOPSIS  
touch [OPTION]... FILE...  
  
DESCRIPTION  
Update the access and modification times of each FILE to the current  
time.  
  
A FILE argument that does not exist is created empty, unless -c or -h  
is supplied.  
  
A FILE argument string of - is handled specially and causes touch to  
change the times of the file associated with standard output.  
  
Mandatory arguments to long options are mandatory for short options  
too.  
  
-a change only the access time  
  
-c, --no-create  
do not create any files  
  
-d, --date=STRING
```

6. **date** :- Displays the current system date and time

```
(shrutirane@shrutirane) ~-/Desktop/s12
$ date
Wednesday 11 December 2025 11:16:02 AM IST
```

7. **free** :- Displays information about system memory usage (RAM and swap)

```
(shrutirane@shrutirane) ~-/Desktop/s12
$ free
              total        used        free      shared  buff/cache   availa
ble
Mem:           2018576      801376      543664         9028       861468      1215
300
Swap:          2093100           12       2091088
```

8. **time** :- Shows how long a command takes to execute

```
(shrutirane@shrutirane) ~-/Desktop/s12
$ time

real    2921.93s
user    0.93s
sys     0.62s
cpu      0%

real    2921.93s
user    0.65s
sys     0.17s
cpu      0%

(shrutirane@shrutirane) ~-/Desktop/s12
```

9. **mkdir** :- Creates a new directory

```
(shrutirane@shrutirane) ~-/Desktop
$ mkdir shruti

(shrutirane@shrutirane) ~-/Desktop
$ ls
shruti
```

10. **rmdir** :- Deletes an empty directory

```
(shrutirane@shrutirane) ~-/Desktop
$ mkdir shruti

(shrutirane@shrutirane) ~-/Desktop
$ ls
shruti

(shrutirane@shrutirane) ~-/Desktop
$ rmdir shruti

(shrutirane@shrutirane) ~-/Desktop
$ ls
```

11. `rm` :- Deletes a file

```
(shrutirane@shrutirane) [~/Desktop]
$ touch a.txt
$ ls
a.txt  Documents  rrr.txt  sll  sss.txt
(shrutirane@shrutirane) [~/Desktop]
$ rm a.txt
(shrutirane@shrutirane) [~/Desktop]
$ ls
Documents  rrr.txt  sll  sss.txt
(shrutirane@shrutirane) [~/Desktop]
$
```

12. `cp` :- Copies files or directories

```
(shrutirane@shrutirane) [~/Desktop]
$ touch a.txt
(shrutirane@shrutirane) [~/Desktop]
$ touch b.txt
(shrutirane@shrutirane) [~/Desktop]
$ echo hello > a.txt
(shrutirane@shrutirane) [~/Desktop]
$ cat a.txt
hello
(shrutirane@shrutirane) [~/Desktop]
$ cp a.txt b.txt
(shrutirane@shrutirane) [~/Desktop]
$ cat b.txt
hello
(shrutirane@shrutirane) [~/Desktop]
$
```

13. `mv` :- Moves or renames files

```
(shrutirane@shrutirane) [~]
$ ls
Desktop  Downloads  Pictures  sample  Videos
Documents  Music  Public  Templates
(shrutirane@shrutirane) [~]
$ mv sample example
(shrutirane@shrutirane) [~]
$ ls
Desktop  Downloads  Music  Public  Videos
Documents  example  Pictures  Templates
```

14. `clear` :- Clears the terminal screen

```
shrutirane@shrutirane ~  
Session Actions Edit View Help  
[shrutirane@shrutirane]~  
$ pwd  
/home/shrutirane  
[shrutirane@shrutirane]~  
$ clear
```

15. **whoami** :- Displays the username of the currently logged-in user.

```
[shrutirane@shrutirane]~  
$ whoami  
shrutirane
```

16. **history** :- Shows the list of previously executed commands

```
shrutirane@shrutirane ~  
Session Actions Edit View Help  
[shrutirane@shrutirane]~  
$ history  
1 su  
2 pwd  
3 ls  
4 cd  
5 cd desktop  
6 ls  
7 cd Desktop  
8 mkdir shruti  
9 ls  
10 rm -rf shruti  
11 ls  
12 find Documents  
13 mkdir abc  
14 mkdir s12  
15 find s12  
16 mv abc Documents  
17 cd Documents  
18 ls  
19 cd ~  
20 cd Documents  
21 ls  
22 ls  
23 cd ~  
24 ls  
25 cd Documents  
26 cd Desktop
```

17. **watch** :- Runs a command repeatedly at regular intervals and shows the output

```
[shrutirane@shrutirane]~/Desktop/s12  
$ watch ls
```

```
shrutirane@shrutirane: ~/Desktop/s12
Session Actions Edit View Help
q, --quit <cycles>      exit when output from command does not change
-n, --interval <secs>  seconds to wait between updates
-p, --precise           attempt run command in precise intervals
-r, --no-rerun         do not rerun program on window resize
-t, --no-title         turn off header
-w, --no-wrap          turn off line wrapping
-x, --exec             pass command to exec instead of "sh -c"

-h, --help            display this help and exit
-v, --version         output version information and exit

for more details see watch(1).
```

18. `cd ..` :- Moves to the parent directory

```
(shrutirane@shrutirane)~/Desktop/
$ cd s12
(shrutirane@shrutirane)~/Desktop/s12/
$ cd ..
(shrutirane@shrutirane)~/Desktop/
$
```

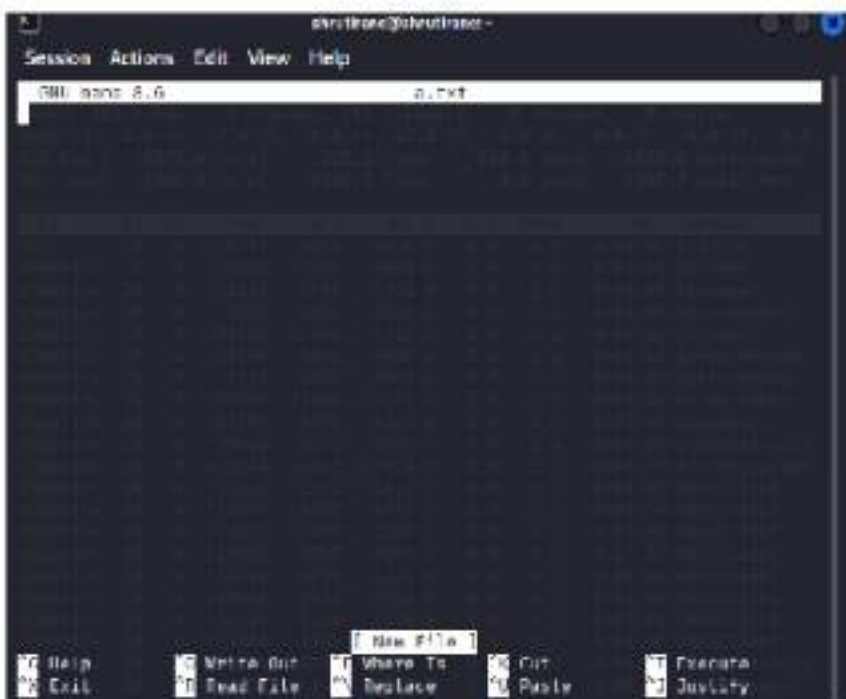
19. `cat` :- Displays the contents of a file, or combines files

```
(shrutirane@shrutirane)~/Desktop/
$ echo hello > a.txt
(shrutirane@shrutirane)~/Desktop/
$ cat a.txt
hello
```

20. `echo` :- Displays a line of text or a variable value on the terminal

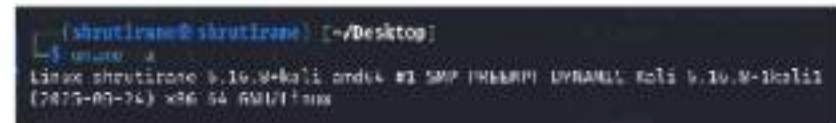
```
(shrutirane@shrutirane)~/Desktop/
$ echo hello > a.txt
(shrutirane@shrutirane)~/Desktop/
$ cat a.txt
hello
```

21. `nano`:- is a simple, user-friendly text editor used in the Linux terminal to create and edit text files.



```
shrutirane@shrutirane:~$ uname -a
Linux shrutirane 5.10.0-kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 5.10.0-kali1 (2020-09-24) x86_64 GNU/Linux
```

22. `uname -a` :- Displays system information



```
(shrutirane@shrutirane) [~/Desktop]
$ whoami
Linux shrutirane 5.10.0-kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 5.10.0-kali1 (2020-09-24) x86_64 GNU/Linux
```

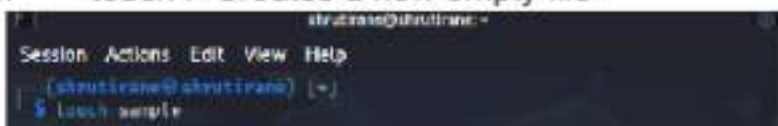
23. `whoami` :- Shows the current user



```
(shrutirane@shrutirane) [~]
$ touch
shrutirane

(shrutirane@shrutirane) [~]
$
```

24. `touch` :- Creates a new empty file



```
shrutirane@shrutirane:~$ touch sample
```


25. ip a :- Displays IP addresses and network interfaces

```
shrutirane@shrutirane:~/Desktop|
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    link/ether 08:00:27:df:fa:a1 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 85544sec preferred_lft 85544sec
    inet6 fd17:625c:f037:2:bd5f:7ba2:1962:9a00/64 scope global temporary dyna
        valid_lft 86271sec preferred_lft 14271sec
    inet6 fd17:625c:f037:2:a00:27ff:fedf:faa1/64 scope global dynamic mgtapi
        valid_lft 86271sec preferred_lft 14271sec
    inet6 fe80::a00:27ff:fedf:faa1/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
shrutirane@shrutirane:~/Desktop|
```

26. ifconfig :- Shows network interface information

```
shrutirane@shrutirane:~/Desktop|
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fedf:faa1 prefixlen 64 scopeid 0<link>
    inet6 fd17:625c:f037:2:bd5f:7ba2:1962:9a00 prefixlen 64 scopeid 0<glo
    global>
    inet6 fd17:625c:f037:2:a00:27ff:fedf:faa1 prefixlen 64 scopeid 0<glo
    global>
    ether 08:00:27:df:fa:a1 txqueuelen 1000 (Ethernet)
    RX packets 1261 bytes 177673 (173.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1416 bytes 145320 (141.9 KiB)
    TX errors 0 dropped 74 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<localhost>
    loop txqueuelen 1000 (local loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```


27. ping :- Tests network connectivity

```
shrutirane@shrutirane: ~/Desktop
$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 64(84) bytes of data:
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=0.045 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.042 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.068 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.041 ms
64 bytes from 10.0.2.15: icmp_seq=5 ttl=64 time=0.038 ms
64 bytes from 10.0.2.15: icmp_seq=6 ttl=64 time=0.038 ms
64 bytes from 10.0.2.15: icmp_seq=7 ttl=64 time=0.038 ms
64 bytes from 10.0.2.15: icmp_seq=8 ttl=64 time=0.039 ms
64 bytes from 10.0.2.15: icmp_seq=9 ttl=64 time=0.043 ms
64 bytes from 10.0.2.15: icmp_seq=10 ttl=64 time=0.041 ms
64 bytes from 10.0.2.15: icmp_seq=11 ttl=64 time=0.047 ms
64 bytes from 10.0.2.15: icmp_seq=12 ttl=64 time=0.049 ms
```

28. netstat :- Displays open and listening ports

```
shrutirane@shrutirane: ~/Desktop/s12
Session Actions Edit View Help

shrutirane@shrutirane:~/Desktop/s12
$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 10.0.2.15:bootpc       10.0.2.2:bootps        ESTABLISH
ED

Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State       I-Node      Path
unix 3      [ ]         STREAM     CONNECTED   11123       /run/user/1000/bus
unix 3      [ ]         STREAM     CONNECTED   10993       /run/user/1000/bus
unix 3      [ ]         STREAM     CONNECTED   10960       /run/user/1000/bus
unix 2      [ ]         DGRAM      CONNECTED   9776        /run/dbus/system_b
unix 3      [ ]         STREAM     CONNECTED   11709       /run/dbus/system_b
unix 3      [ ]         STREAM     CONNECTED   10131       /run/user/1000/at-
spi/bus_0
unix 3      [ ]         STREAM     CONNECTED   10580       /run/user/1000/bus
unix 3      [ ]         STREAM     CONNECTED   11904
unix 3      [ ]         STREAM     CONNECTED   11827
unix 3      [ ]         STREAM     CONNECTED   7093
unix 3      [ ]         STREAM     CONNECTED   12089
unix 3      [ ]         STREAM     CONNECTED   11499
unix 3      [ ]         DGRAM      CONNECTED   10162       /tmp/.X11-unix/X0
unix 3      [ ]         DGRAM      CONNECTED   9037
unix 3      [ ]         STREAM     CONNECTED   11236       /tmp/.X11-unix/X0
```

29. ss -tln :- Shows active network connections

```
Session Actions Edit View Help

shrutirane@shrutirane: ~
$ ss
```

```

shrutirane@shrutirane: ~
Session Actions Edit View Help
+ str LSIAA 0 0 A
10905 + 10906
+ str FSTAA 0 2 A
10919 + 10920
+ str LSIAA 0 2 A
9887 + 9888
+ str LSIAA 0 2 jcd193cb/3ac9/4a8/bus/systemd/bus-system
9847 + 9848
+ str FSTAA 0 2 2e70d9dhd021a4774/bus/systemd-logged/system
7010 + 7011
+ str LSIAA 0 0 8197e/208be4e42f/bus/systemd/bus-api-system
1758 + 1759
+ str FSTAA 0 2 6744c2f875e415677/bus/systemd/bus-api-system
10111 + 10112
10066 IMCONA 0 3 ++
ipw6 Lupa
+ str LSIAA 0 2 10.0.2.15:10000
10076 10.0.2.2:10000

```

30. **iwconfig** :- Displays wireless network information

```

shrutirane@shrutirane: ~
$ iwconfig
lo        no wireless extensions.

eth0      no wireless extensions.

```

31. **ping** – Checks network connection

```

shrutirane@shrutirane: ~
$ ping google.com
PING google.com (142.251.43.14) 56(84) bytes of data:
64 bytes from prbamb-bb-in-f14.1e100.net (142.251.43.14): icmp_seq=1 ttl=250
time=73.4 ms
64 bytes from prbamb-bb-in-f14.1e100.net (142.251.43.14): icmp_seq=2 ttl=250
time=82.0 ms
64 bytes from prbamb-bb-in-f14.1e100.net (142.251.43.14): icmp_seq=3 ttl=250
time=72.6 ms
64 bytes from prbamb-bb-in-f14.1e100.net (142.251.43.14): icmp_seq=4 ttl=250
time=82.0 ms
64 bytes from prbamb-bb-in-f14.1e100.net (142.251.43.14): icmp_seq=5 ttl=250
time=71.8 ms
64 bytes from prbamb-bb-in-f14.1e100.net (142.251.43.14): icmp_seq=6 ttl=250
time=71.6 ms

```

32. **id** :- Displays user ID (UID) and group ID (GID)

```

shrutirane@shrutirane: ~/Desktop
$ id
uid=1000(shrutirane) gid=1000(shrutirane) groups=1000(shrutirane),4(adm),10(da
lout),24(cdrom),25(floppy),27(cdrom),29(audio),30(dip),44(video),46(plugdev)
,100(users),101(sandev),103(scanner),116(bluetooth),121(lpadmin),130(wireles
s),110(vboxsf),111(kuboxer)

```

33. nslookup :- DNS lookup tool used to find the IP address of a domain name

```
(shrutirane@shrutirane)~$ nslookup
> google.com
Server:      192.168.1.1
Address:     192.168.1.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.207.238
Name:   google.com
Address: 2a04:6800:1400::103c
```

34. cal :- Displays the calendar

```
(shrutirane@shrutirane)~$ cal
December 2023
Su Mo Tu We Th Fr Sa
    1  2  3  4  5  6
 7  8  9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30 31
```

35. ping websitename :- Checks network connection to a website

```
(shrutirane@shrutirane)~$ ping google.com
PING google.com (142.251.43.14): 56(84) bytes of data:
64 bytes from prbmb-ba-in-14.1e190.net (142.251.43.14): icmp_seq=1 ttl=255
time=33.4 ms
64 bytes from prbmb-ba-in-14.1e190.net (142.251.43.14): icmp_seq=2 ttl=255
time=33.3 ms
64 bytes from prbmb-ba-in-14.1e190.net (142.251.43.14): icmp_seq=3 ttl=255
time=33.8 ms
64 bytes from prbmb-ba-in-14.1e190.net (142.251.43.14): icmp_seq=4 ttl=255
time=33.9 ms
64 bytes from prbmb-ba-in-14.1e190.net (142.251.43.14): icmp_seq=5 ttl=255
time=33.8 ms
64 bytes from prbmb-ba-in-14.1e190.net (142.251.43.14): icmp_seq=6 ttl=255
time=33.4 ms
```

36. sudo adduser username :- Adds a new user to the system.

```
(shrutirane@shrutirane)~$ sudo adduser abc
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for abc
Enter the new value, or press ENTER for the default
  Full Name []: abc
  Room Number []: 1
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
```

37. **--help :-** Shows help information for a command

```
(shrutirane@shrutirane)~$ pgrep --help
+o, --oldest select least recently started
+O, --older <seconds> select those older than seconds
+P, --parent <PPID, ...> match only child processes of the given parent
+e, --session <SID, ...> match session IDs
+s, --signal <sig> signal to send (either number or name)
+T, --terminal <tty, ...> match by controlling terminal
+x, --uid <UID, ...> match by effective UIDs
+u, --uid <ID, ...> match by real IDs
+X, --exact match exactly with the command name
+P, --pidfile <file> read PIDs from file
+L, --legidfile <file> fail if PID file is not locked
+r, --monstates <states> match monstates {D,S,Z,...}
+X, --ignore-ancestor <group <grp, ...> exclude our ancestors from results
+u <PID> match by egroup w/ names
match the processes that belong to the same namespace as <pid>
--nolist <ns, ...> list which namespaces will be considered for the --ns option.
Available namespaces: ipc, mnt, net, pid, uts, etc.

+h, --help display this help and exit
+V, --version output version information and exit

For more details see pgrep(1).

(shrutirane@shrutirane)~$
```

38. **sudo apt update :-** Updates the package list

```
(shrutirane@shrutirane)~$ sudo apt update
[sudo] password for shrutirane:
Get:1 http://mirrors.etsi.network/kali kali-rolling InRelease [34.6 kB]
Get:2 http://mirrors.etsi.network/kali kali-rolling/main amd64 Packages [10.9 MB]
Get:3 http://mirrors.etsi.network/kali kali-rolling/main amd64 Sources [deb] [52.5 kB]
100% [2 Packages done] 3.0 B/s [3 Contents-archives 3.0/52.5 KB 0%] 3,483 B/s 310
```

39. **apt upgrade :-** Upgrades installed packages


```
stratix@stratix: ~  
Session Actions Edit View Help  
RPM/Tab:  
connix: info:connix:connix python2-wit  
git: info:git:git python2-metasploit  
kali-linux-default: info:kali-linux-default:kali  
kali-linux-headless: info:kali-linux-headless:kali  
metasploit-framework: info:metasploit-framework:metasploit-framework  
Summary:  
Upgrading: 0, Installing: 0, Removing: 17, Not upgrading: 794  
Free space: 599 MB  
Continue? [Y/n] y  
(Reading database ... 422838 files and directories currently installed.)  
Removing kali-linux-default (2025.4.11) ...  
Removing kali-linux-headless (2025.4.11) ...  
Removing connix (4.0-0kali1) ...  
Removing git (0.9.1-0kali1) ...  
Removing kali-tools-topio (2025.4.11) ...  
Removing python2-pyexpintdb (0.2.102-0kali1) ...  
Removing python2-wit (0.2.45-1) ...  
Removing metasploit (0.4.5-0kali1) ...  
Removing net (0.0.3+git10241021-0kali1) ...  
Removing anisome-magic (0.12-0kali1) ...  
Removing metasploit-framework (0.4.99-0kali1) ...  
Removing git (1:2.51.0-1) ...  
Processing triggers for libc-bin (2.41-12) ...  
Processing triggers for man-db (2.13.1-1) ...  
Progress: 100% |
```